

MSXPi Quick Start Guide

Version 1.6

Real hardware (MSXPi interface + Raspberry Pi) and openMSX. Everything else is in the Users and Developers Guide.

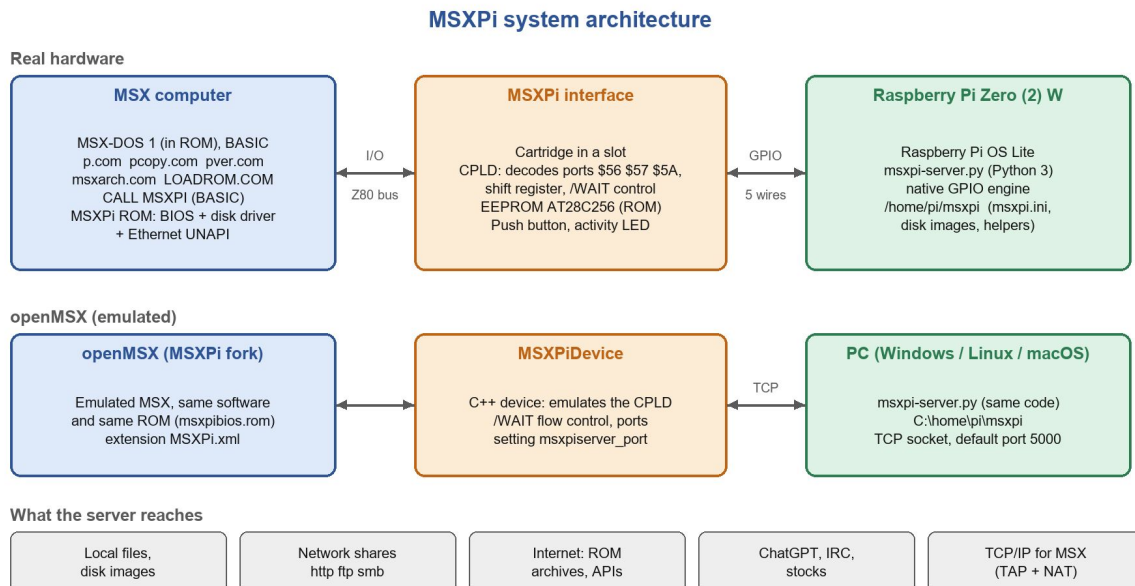


Figure 1: What you are setting up

Read this first: Version 1.6 breaks compatibility with earlier ROMs and servers. Upgrade the ROM, the msxpi-server, the client commands (.COM files) and, on real hardware, the CPLD firmware together. A v1.6 ROM on older CPLD firmware still works, only slower.

Pick your path: **Part 1** for a real MSX with an MSXPi interface, **Part 2** for openMSX with no hardware. Each step ends with a check - do not go on until the check passes.

Commands typed on the MSX are shown as `C:\>` lines or in a grey box; the same commands work on hardware and in openMSX.

1 Part 1 - Real hardware

1.1 What you need

- The MSXPi interface with an AT28C256 EEPROM. Build notes: [documents/MSXPi Assembly Instructions.txt](#).
- A Raspberry Pi Zero W or Zero 2 W (a Pico does not work) with its 40-pin header on the side that matches your PCB, and a micro SD card of 8 GB or more.
- An MSX with a disk or SD interface (MSX-DOS 1, MSX-DOS 2 or Nextor) and a way to copy files to it from a PC. MSXPi's own ROM contains MSX-DOS 1 only: Nextor or MSX-DOS 2 must come from that other disk interface, which MSXPi coexists with - the best setup.
- Only if the CPLD is not programmed yet: an Altera USB-Blaster and Quartus 13.0 SP1 on Windows.
- The files in this repository: [software/target](#) (programs and ROM), [hardware/CPLD_Project](#) (firmware).

1.2 Step 1 - Program the CPLD (only if new or older than v1.6)

1. Plug the interface into the MSX, connect the USB-Blaster to the JTAG connector, switch the MSX on.
2. In Quartus: Tools, Programmer, mode JTAG, Add File, choose the [.pof](#) for your board, tick Program/Configure and Verify, Start.

Board	File in hardware/CPLD_Project
V1.3 Rev.1 (built for v1.3 to v1.5)	MSXPi_v1.6.pof
V1.1 Rev.1, V1.2.1b	v1.1/MSXPi_v1.6_wait.pof
v0.8.2 / v1.0 (PCB v0.7 Rev.7)	v0.8.2/MSXPi_v1.6_polled.pof

Check: in BASIC, [PRINT INP\(&H57\)](#) must not print 254 or 255. Older boards need extra settings later (see the Legacy Support document).

1.3 Step 2 - Write the ROM to the EEPROM

1. Copy [at28c256.com](#) and [msxpibios.rom](#) from [software/target](#) to an MSX disk. The interface must be plugged in; the Pi is not needed yet.
2. If the interface already boots a ROM, remove its enable jumper (SLTSL) first.
3. Boot MSX-DOS and list the slots: [at28c256 /i](#). Write down the slot of the interface.
4. Write the ROM (slot 2 in this example): [at28c256 /s 2 msxpibios.rom](#). Add [/r](#) to write slowly if your chip fails. The chip is write-protected afterwards.
5. Power off and set the jumpers. **With another disk interface (recommended): remove SLTSL.** To boot MSX-DOS 1 from a disk image on the Pi: close SLTSL and select the bank with the MSXPi ROM.

Check: with the enable jumper closed, **CALL MSXPIVER** in BASIC prints the version.

1.4 Step 3 - Prepare the Raspberry Pi

Easiest: the ready-made SD card image. Download <https://tinyurl.com/MSXPi-SDCardV3>, unzip it with 7-Zip, write it with Win32DiskImager or Raspberry Pi Imager ("Use custom").

From scratch: write "Raspberry Pi OS Lite (32-bit)" with Raspberry Pi Imager (user **pi**, SSH on, your WiFi entered), boot the Pi, log in and run:

```
wget https://tinyurl.com/MSXPi-Setup
chmod 755 MSXPi-Setup
bash ./MSXPi-Setup
```

The script installs Python 3 with the modules **requests**, **fs** and **pyfatfs**, the packages for archives and audio, the MSXPi home directory **/home/pi/msxpi** (server, **msxpi.ini**, disk images, helpers), the service that starts the server at boot, and reboots. It can take over an hour on a Pi Zero.

In **msxpi.ini** a board **without** the shutdown push-button needs **var RPI_SHUTDOWN=none**. Optional keys: OPENAIKEY (ChatGPT), and the stock quote keys.

Insert the card, plug the Pi into the interface **component side facing you**, switch the MSX on. The interface LED lights when the server runs.

1.5 Step 4 - Copy the programs to the MSX

Copy everything in **software/target** (at least **p.com**, **pcopy.com**, **pver.com**, **msxarch.com**, **msxarch.ini**, **LOADROM.COM**, **INL.CFG**) to your MSX disk or SD card.

```
pver
```

Check: **pver** prints the board, the CPLD build, **/WAIT: yes (burst)** and the server version.

"Connection error"? The Pi may still be booting (first boot: up to three minutes), or the versions do not match. Press ESC, wait five seconds, try again.

1.6 Step 5 - WiFi and network

```
p set WIFISSID Your Wifi Name
p set WIFIPWD YourWifiPassword
p wifi set
p reboot
```

Then, for TCP/IP programs (the Ethernet UNAPI is in the ROM; do not run RAMHELPR):

```
MSR I          (MSX-DOS 1 only)
INL I
HOST GOOGLE.COM
```

Check: **HOST** prints an address. If not: **p netreset**. "DNS not found" means the resolver: use **INL IP P <address>**.

1.7 Step 6 - Have fun

```
p dir /home/pi/msxpi
pcopy m:pver.com
msxarch
```

p chatgpt Tell me a fun fact about MSX

Update later with **msxpiupd.bat** (MSX files), the setup script (server) and **at28c256** (ROM). Run **`p shut`** before switching the MSX off.

2 Part 2 - openMSX

2.1 What you need

- **openMSX with the MSXPi device.** The official openMSX is <https://github.com/openMSX/openMSX>; the MSXPi device is in the fork - download the newest release for your system from <https://github.com/costarc/openMSX/releases> (read its release notes).
- **Python 3.9 or newer**, with `python -m pip install requests fs pyfatfs`.
- **7-Zip** on the PATH (zip, lzh, pma, 7z archives).
- The MSXPi home directory: `/home/pi/msxpi` (on Windows `C:\home\pi\msxpi`).
- The files from this repository: the server in `software/Server/Python/src`, the programs in `software/target`.

Windows shortcut: run `software/Server/Shell/msxpi-windows-setup.ps1` (right-click, Run with PowerShell). It installs Python, 7-Zip, the modules, the home directory, openMSX with MSXPi, the TAP driver and a launcher with a desktop shortcut. Then skip to step 4.

2.2 Step 1 - Create the home directory

Put in it: `msxpi-server.py`, `mapper_detect.py`, `msxpi_eth.py`, `msxpi.ini` (copy `msxpi-JumperLeft.ini` and rename it) and a `disks` folder with `msxpiboot.dsk` and `tools.dsk`. On Windows create `C:\home\pi\msxpi` and start the server from drive C: - the server path resolves on the current drive.

2.3 Step 2 - Start the server

```
cd C:\home\pi\msxpi
python msxpi-server.py
```

It listens on TCP port 5000 and prints every command it receives. Leave it running.

2.4 Step 3 - Prepare a disk folder

Make a folder `FloppyA`. Copy `MSXDOS.SYS` and `COMMAND.COM` (from `software/target/disks`) and all the MSXPi programs from `software/target` into it.

2.5 Step 4 - Start openMSX

```
| openmsx -machine Panasonic_FS-A1WSX -ext MSXPi -diska FloppyA
```

Check: at the DOS prompt type `pver`. It should show `/WAIT: yes (burst)` and the server version. "Connection error": the server is not running, the ports differ, or the versions do not match.

Optional - another TCP port: in the openMSX console (F10) type `set msxpiserver_port 5001` and change the `PORT` value at the top of `msxpi-server.py` to match. Default: 5000.

2.6 Step 5 - Network in the emulator (optional)

Windows: install the OpenVPN TAP driver, run `powershell -ExecutionPolicy Bypass -File msxpi-tcpip-setup.ps1` as **Administrator** (again after every reboot), start the server, then on the MSX `MSR I` (MSX-DOS 1 only), `INL I`, `HOST GOOGLE.COM`. If the server prints "falling back to MockLink", something else has the adapter open.

3 Cheat sheet

Command	Does
p ver	Board, CPLD, /WAIT and server version
p dir [path]	List Pi, network or r1: r2: m: locations
p cd [path]	Set / show the current path
pcopy src [dest]	Copy Pi/network to MSX (/z unpacks); pcopy a:f /path copies to the Pi
p run cmd	Run a command on the Pi (: : for a pipe)
p set NAME value	Change msxpi.ini variables (WIFISSID, OPENAIKEY, ...)
p wifi [set], p wlanreset, p netreset	WiFi, WiFi reset, rebuild the network for the MSX
p date	Set the MSX clock from the Pi
p reboot / shut / restart	Reboot the Pi / shut it down / restart the server
msxarch	Browse ROM repositories and start a game
LOADROM name /N	Load a ROM from the network
p chatgpt text	Ask ChatGPT (needs OPENAIKEY)
ESC	Abort; wait 5 seconds before the next command

4 When something goes wrong

Symptom	Try
"Connection error" from pver or p	Pi still booting (wait 3 min on the first boot); ESC, wait 5 s, retry; run the server by hand and read its messages; check that ROM, server and .COM files are all v1.6
pver: Board: no MSXPi found	CPLD not programmed, or the interface is not seated; check PRINT INP(&H57)
pver: /WAIT: no (polled)	Older CPLD firmware or a board without /WAIT: works, but slower
Pi reboots by itself (older boards)	Add var RPI_SHUTDOWN=none to msxpi.ini
INL: No mapper support routines found	RAMHELPR was run before MSR: cold boot, then MSR and INL only
HOST: DNS not found	INL IP P <resolver address> ; run p netreset
MSX-DOS 1 does not boot from the Pi	Wait for the Pi to boot; ESC to skip to BASIC; replace msxpiboot.dsk
openMSX: nothing happens	Start msxpi-server.py first; port must match msxpiserver_port ; use the fork build of openMSX

More: **README.md** in the repository, the **MSXPi Users and Developers Guide**, **documents/Legacy Support.odt** for older boards, and the two Raspberry Pi tutorials (Easy Way and Hard Way; English and Portuguese). Issues: <https://github.com/costarc/MSXPi/issues>